

# Simulation and Reconstruction of $e^+$ and $e^-$ Trajectories in a Time Projection Chamber with Orthogonal Fields

9th International MPGD Conference

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Preprint: [arXiv:2606.18821](https://arxiv.org/abs/2606.18821) [1]

August 31, 2026

Funding: GAČR GA21-21801S

Computational resources were provided by the e-INFRA CZ project (ID:90254),  
supported by the Ministry of Education, Youth and Sports of the Czech Republic.



# Outline

- 1 Motivation
- 2 Track Simulation
- 3 Track Reconstruction
- 4 Energy Reconstruction
- 5 Summary & Future



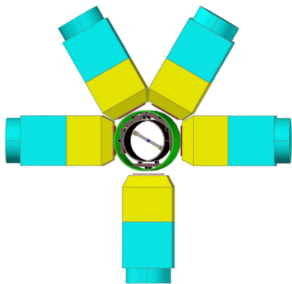
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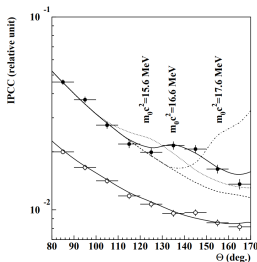
# Motivation: ATOMKI Measurements

- Measurement of anomalies in the angular correlation of an electron-positron pair internally produced in excited  $^8\text{Be}$  [2, 3],  $^4\text{He}$  [4, 5] and  $^{12}\text{C}$  [6]
- Angular resolution  $\sim 7^\circ$ , energy resolution 4.5% at 17.6 MeV

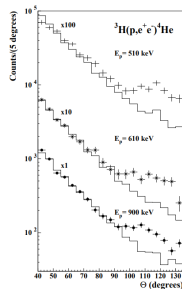


ATOMKI spectrometer. [7]

Beam pipe (black), MWPC,  $\Delta E$  det. (red),  $E$  scintillators (yellow), light guides (blue)



$^8\text{Be}$ ,  $e^+e^-$  pair angular correlation. [2]



$^4\text{He}$ ,  $e^+e^-$  pair angular correlation. [4]

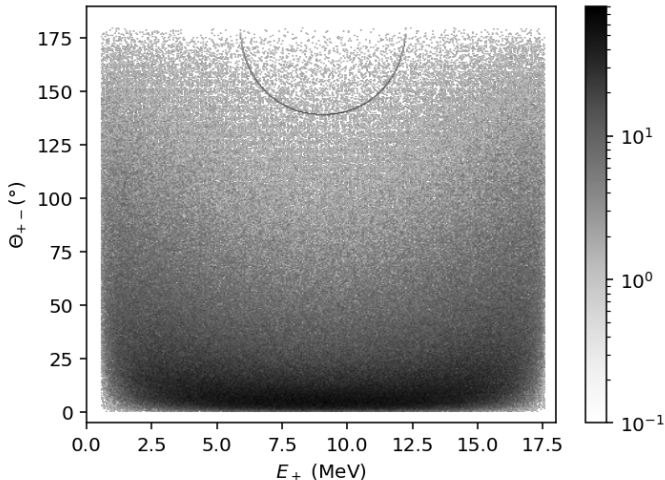


# ATOMKI Anomaly: Other Experiments, Explanations

- In 2023 excesses in  $\gamma\gamma$  spectra ( $\sim 17$  MeV and  $\sim 38$  MeV) in p+C, d+C and d+Cu collisions at Nuclotron in Dubna [8]
- In 2024  $^8\text{Be}$  anomaly also measured using a different spectrometer in Hanoi [9]
- In 2024 the MEG II experiment found no significant excess in  $^8\text{Be}$  and set a limit for the branching ratio [10]
- In 2025 PADME observed a  $2\sigma$  excess when irradiating a fixed target with a positron beam at  $\sqrt{s} = 16.9$  MeV
- Possible explanations include a new protophobic boson X17 [11]



# X17 Decay Signature in IPC

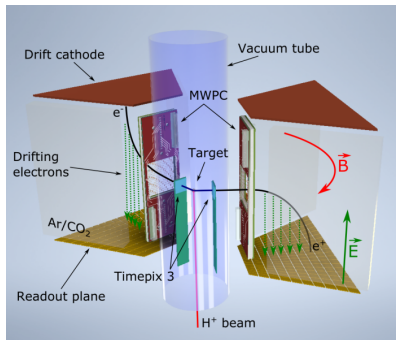


**Geant 4 simulation (ideal case):** Histogram of  $e^+e^-$  pair angle and positron energy produced by standard IPC and by the decay of a hypothetical boson X17 (branching ratio  $3 \cdot 10^{-3}$ ).

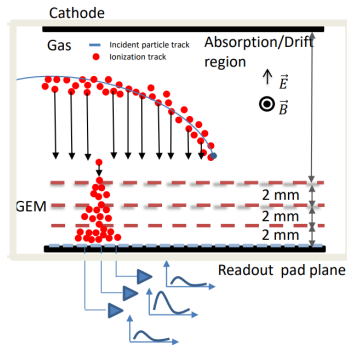


# OFTPC: Detector Configuration

- Time Projection Chamber with Orthogonal Fields (OFTPC) – electric and magnetic field perpendicular (gas mixture Ar/CO<sub>2</sub> – 70/30, 90/10)

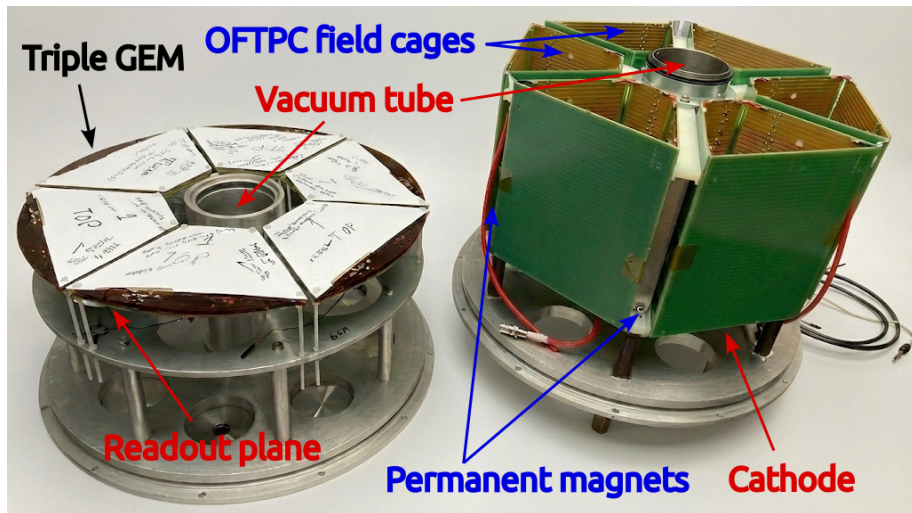


Two out of the six OFTPC chambers. [12]

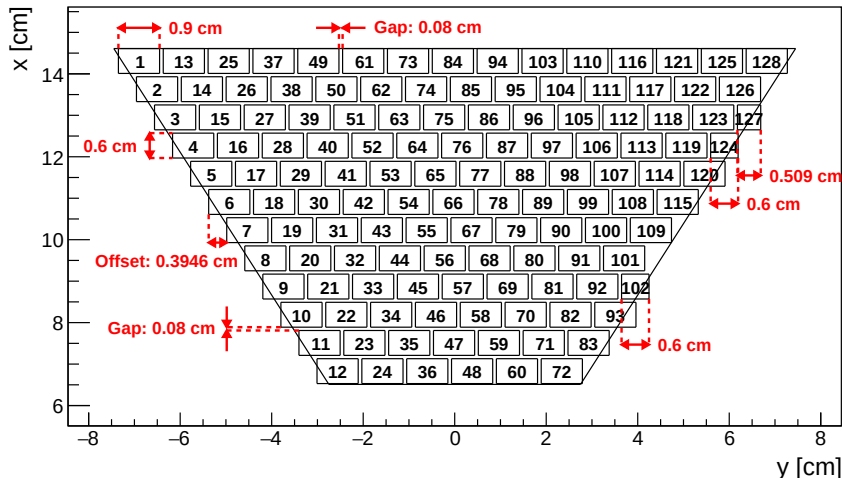


OFTPC with a triple gas electron multiplier (GEM) readout. [12]

# OFTPC: Detector Configuration



# OFTPC: GEM Readout — Pads



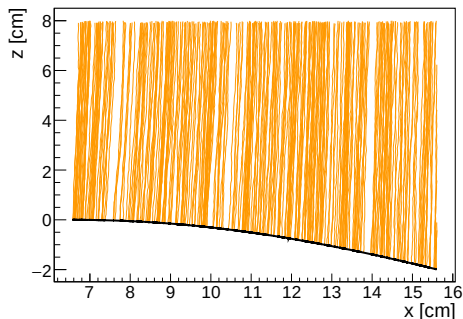
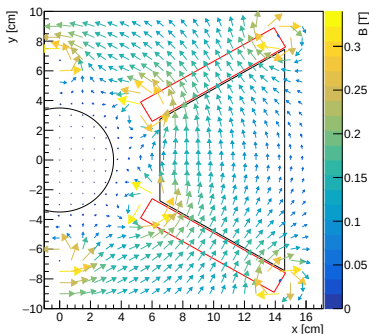
# OFTPC: Reasons for Orthogonal Fields

- No solenoid – permanent magnets used to generate the field
  - Parallel fields difficult to create with permanent magnets
- Space constraints – granularity of the TPC readout limited in order to fit one SAMPA/SRS hybrid in each of the six sectors
  - Parallel fields would bend particles parallel to readout, requiring a much larger number of pads
  - These trajectories would extend to more than one sector, requiring alternative architecture of the detector
- Construction feasibility, cost



# OFTPC: Complications

Inhomogeneous magnetic field (simulated using Maxwell [13])



- The field interferes with the direction of the drift of ionization electrons (position dependent Lorentz angle)
- Curvature of the track is not constant in this field (deviation from a circle)



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# Track Simulation: Initial Approach

- Garfield++ used for track simulation
  - Primary relativistic particle simulated using the HEED program [14]
  - Secondary ionization electrons simulated using microscopic tracking (uses equations of motion)
    - Relatively slow (typically 5-30 CPU hours per track), very precise, especially for small structures.
  - Charge multiplication, delta electrons, attachment, Penning transfer not taken into account
  - HEED does not simulate multiple scattering for primaries
- Five batches of 9702 tracks with different initial parameters simulated on a grid (MetaCentrum [15])
  - Electrons and positrons
  - 11 different energies from 3 MeV to 13 MeV (covers range for  $^8\text{Be}$ )
  - 21 different angles  $\varphi$  and 21 different angles  $\theta$  (next slide)



# Track Simulation: Parameters

Spherical angles  $(\theta, \varphi)$  with respect to  $z$   
 $\theta$  taken from the equatorial plane  $xy$

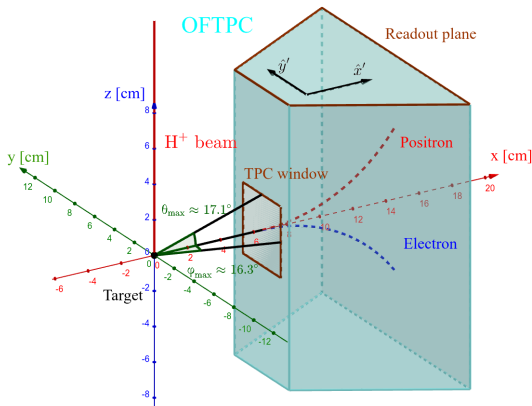


Diagram of the batch simulation parameters:

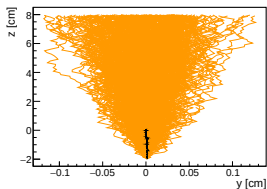
$$\theta \in [-17.1^\circ, 17.1^\circ], \varphi \in [-16.3^\circ, 16.3^\circ], E_{\text{kin.}} \in [3, 13] \text{ MeV.}$$

Made with GeoGebra®.

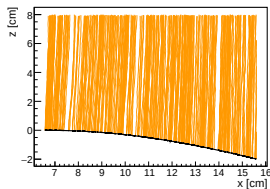


# Simulated Track Example (microscopic tracking)

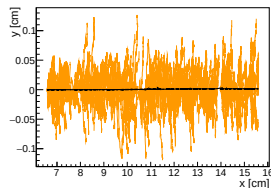
- Electron track with kinetic energy 8 MeV,  $\theta = 0^\circ$  and  $\varphi = 0^\circ$
- Diffusion less than 1.5 mm in both directions



Diffusion front view



Electron drift

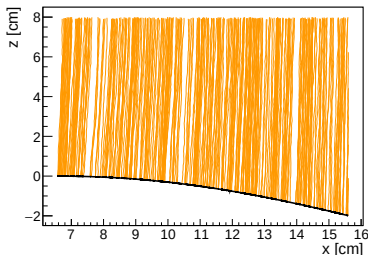


Diffusion top view

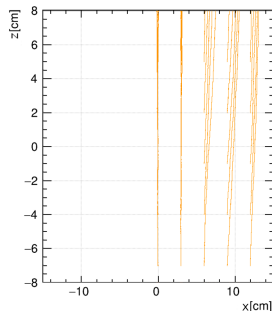


# Ionization Electron Map Simulation

- We want an unambiguous map of the drift of ionization electrons for the reconstruction
- We can use a simulation of evenly spaced electrons
  - Current spacing 5 mm, 100 electrons simulated in each location with 0.1 eV energy in a random direction



Electron drift

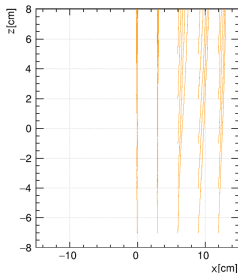


Partial simulation of the map



# Ionization Electron Map Simulation: Principle

- As a result we get an approximation of a mapping from initial coordinates of the electrons  $(x, y, z)$  to the readout coordinates  $(x', y', t)$
- By interpolating we can get the inverse map
- We can use the inverse map to finally create mapping from our discrete readout values (channel number, time) to voxels of the primary track

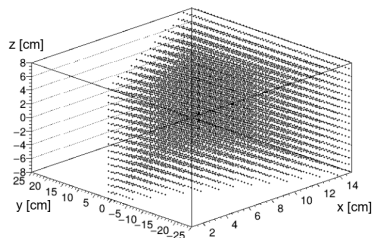


Partial simulation of the map



# Ionization Electron Map Simulation: Visualization

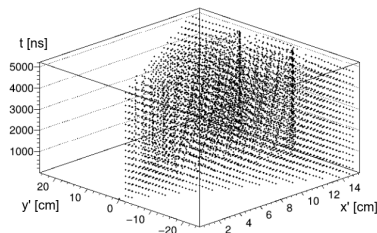
Detector space  $\mathcal{D}$



$\overline{\mathcal{M}}$

$\overline{\mathcal{M}}^{-1}$

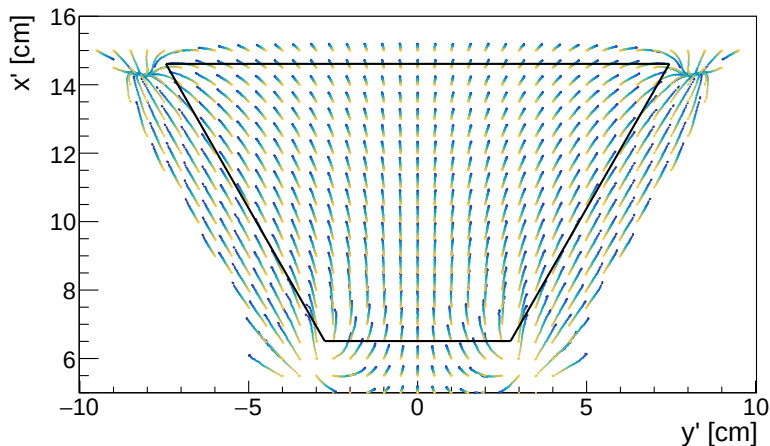
Readout space  $\mathcal{R}$



3D visualization of the simulated mapping  $\overline{\mathcal{M}}$  and the inverse mapping  $\overline{\mathcal{M}}^{-1}$ .



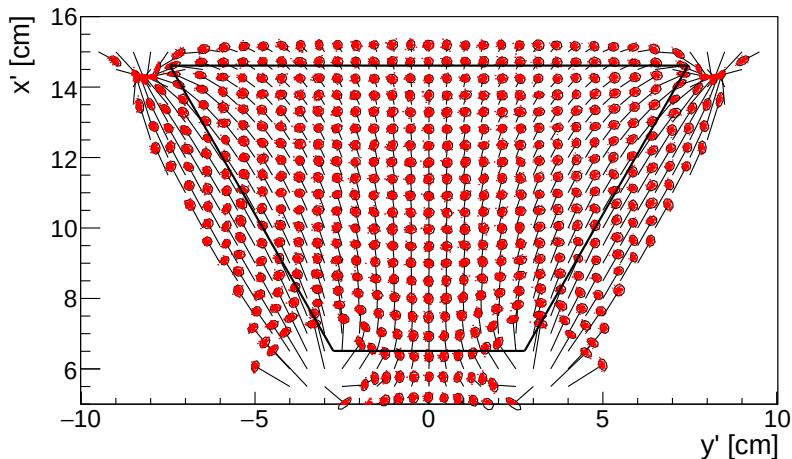
# Ionization Electron Map Simulation: Coordinate Distortion



$x$  and  $y$  coordinate distortion at different  $z$  values (denoted by colors).



# Ionization Electron Map Simulation: Diffusion



Worst case  $x$  and  $y$  coordinate distortion for maximal initial distance from readout. Ellipses with 95 % confidence are shown for each starting position.



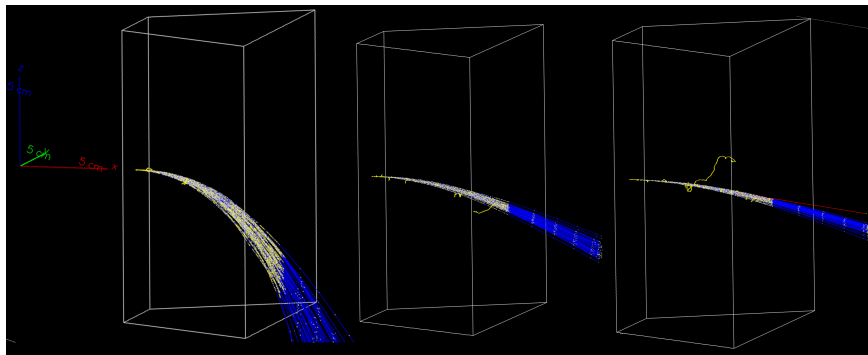


# Track Simulation: Multiple Scattering

- HEED does not account for multiple scattering of relativistic primaries in the gas mixture.
- At relatively low  $e^\pm$  energies of 3–13 MeV, it can significantly distort the trajectory.
- Therefore, we are implementing a Geant4-Garfield++ interface (inspired by Pfeiffer et al. [16]).
- Primary ionization centers are simulated by Geant4, delta electron transport and charge transport by Garfield++.
- Simulation of charge transport can be sped up dramatically by sampling the readout distributions obtained in the map simulation.
- Batches of tracks with special parameters simulated to test the reconstruction, full parameter grid simulation is being implemented.



# Track Simulation: Multiple Scattering



Geant4-Garfield++ interface: Visualization three sets of 50 electron tracks with zero initial angles  $\theta, \varphi$  and energies 3, 8, and 13 MeV.



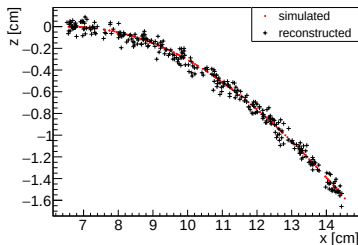
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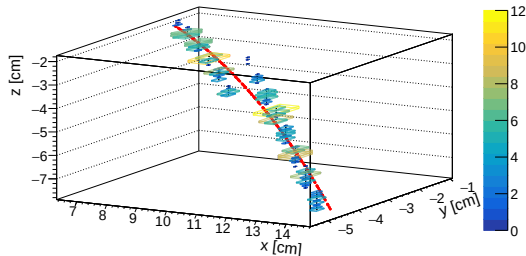


# Track Reconstruction

- At first using only the inverse map (not accounting for readout pads)
- Later simple reconstruction with pads and time bins, counting the number of electrons in each bin



Original and reconstructed  
interaction points on the  
simulated track

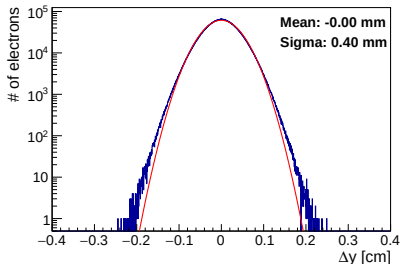
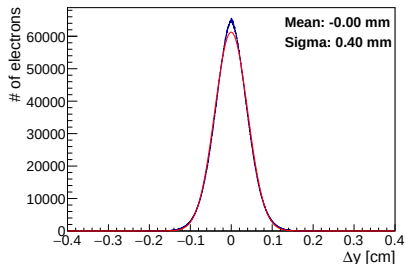
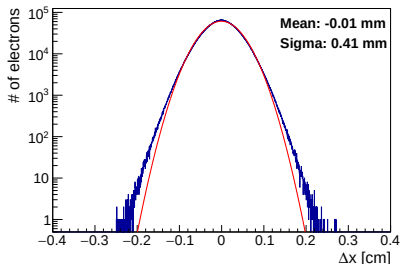
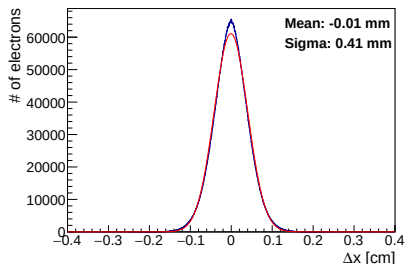


Reconstruction with pads



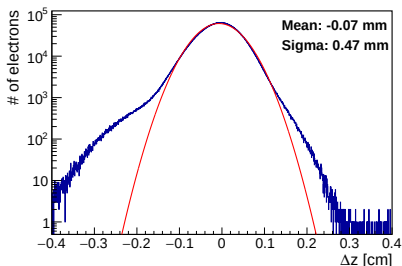
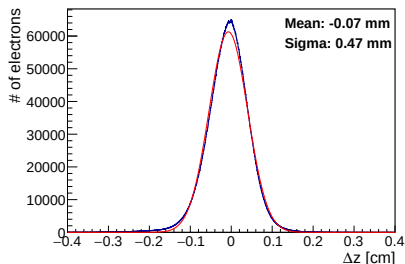
# Ionization Electron Map Reconstruction Residuals ( $x, y$ )

Residuals of init. positions of 14.3 M simul. track ion.  $e^-$  reconstructed using the map



# Ionization Electron Map Reconstruction Residuals ( $z$ )

Residuals of initial positions of 14.3 million simulated track ionization  $e^-$  reconstructed using the map

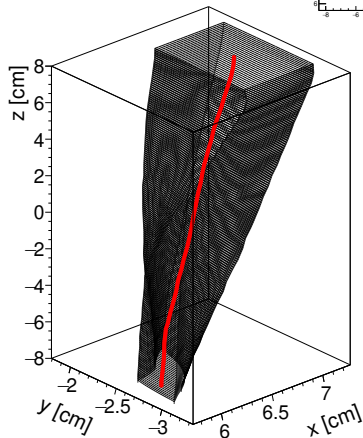


Deviations likely caused by non-zero initial velocity of electrons

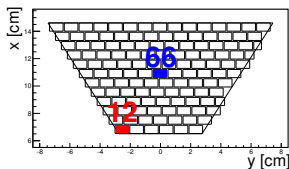


# Inverse Mapping of Pads

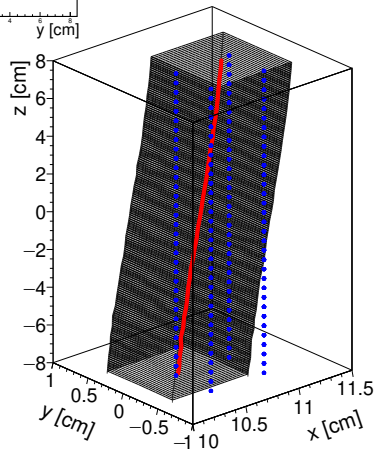
Pad 12  
(near magnet pole)



Inversion by gradient descent



Pad 66  
(near center)



Inversion by interpolation

# Outline

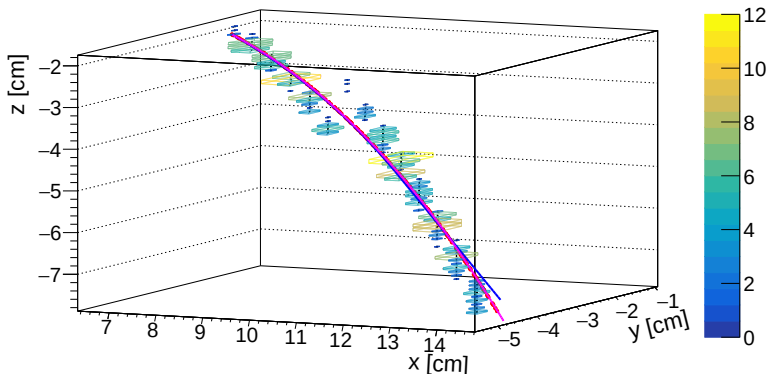
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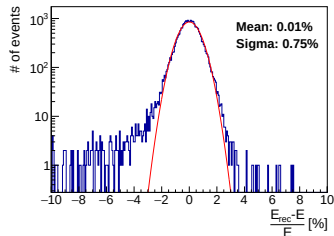
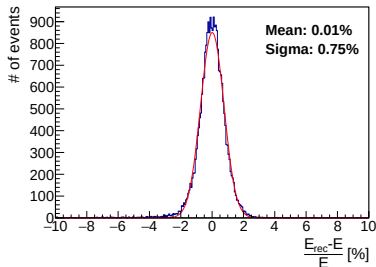
# Energy Reconstruction

- Prefit with circle with smoothly attached lines (blue)
- Kinetic energy fit using 4<sup>th</sup> order Runge-Kutta (magenta)
- Known initial position and direction of the particle assumed
- Apply systematic error correction (obtained from simulation using a 3D linear fit)
- Currently around 0.3 CPU seconds per track

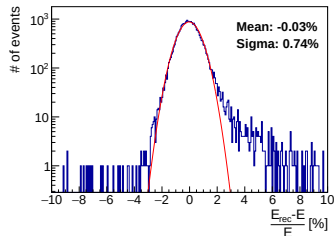
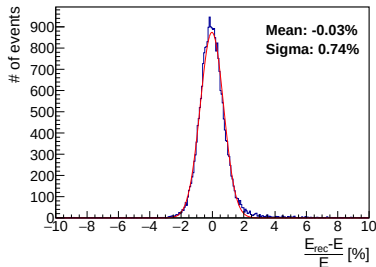


# Energy Reconstruction Precision (overall)

## Electrons



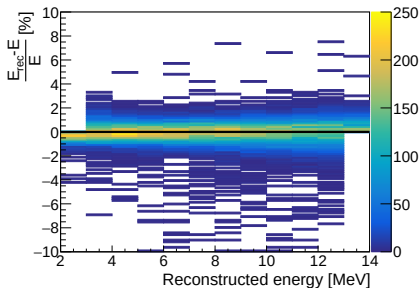
## Positrons



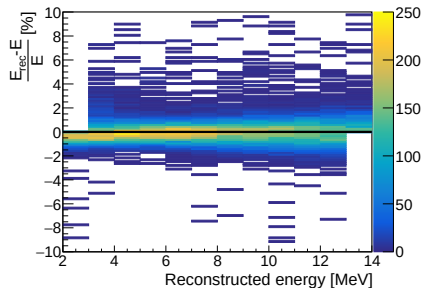
Relative reconstruction deviation of the kinetic energy of electron and positron tracks (ca. 24 000 of each simulated).

# Energy Reconstruction Precision (energy dependence)

## Electrons



## Positrons

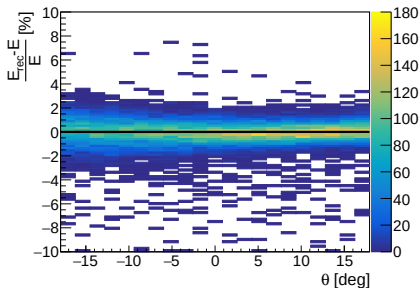


Relative reconstruction deviation of the kinetic energy of electron and positron tracks (ca. 24 000 of each simulated).

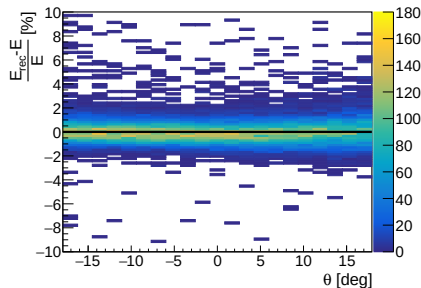


# Energy Reconstruction Precision (theta dependence)

## Electrons



## Positrons

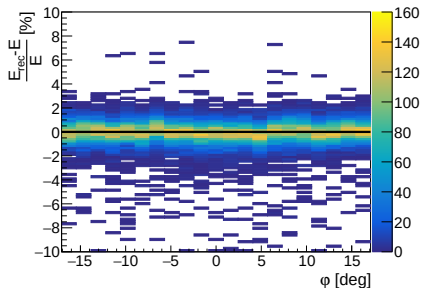


Relative reconstruction deviation of the kinetic energy of electron and positron tracks (ca. 24 000 of each simulated).

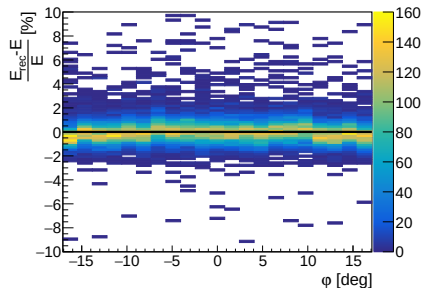


# Energy Reconstruction Precision (phi dependence)

## Electrons



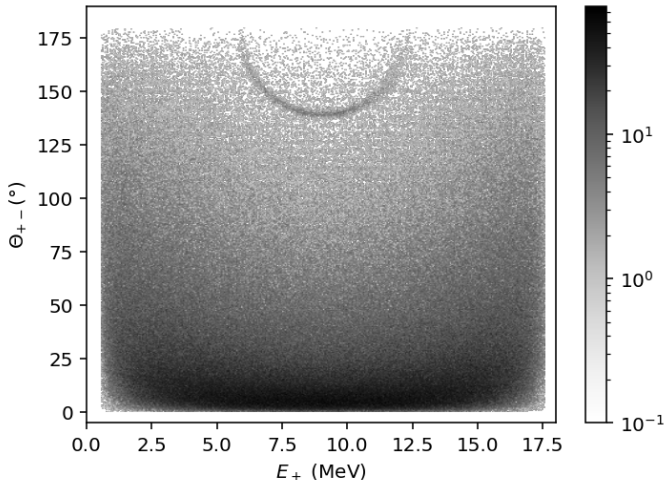
## Positrons



Relative reconstruction deviation of the kinetic energy of electron and positron tracks (ca. 24 000 of each simulated).



# X17 Decay Signature in IPC (with preliminary errors)



**Geant 4 simulation + errors:** Histogram of  $e^+e^-$  pair angle and positron energy produced by standard IPC and by the decay of a hypothetical boson X17 (branching ratio  $3 \cdot 10^{-3}$ ).



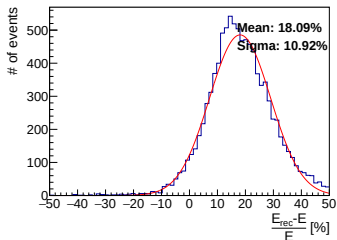
# Effect of multiple scattering (preliminary)

- Preliminary tests show significant impact of multiple scattering on the resolution.
  - Reconstruction of electron tracks with zero  $\theta, \varphi$  and total energy 3, 8 and 13 MeV.
  - Simulation of the full parameter grid is being prepared.
- Delta electrons not taken into account for better comparison with previous simulations.
- Initial position and orientation still fixed in the fit (might increase the error).

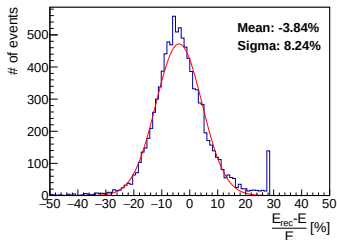


# Effect of multiple scattering (preliminary)

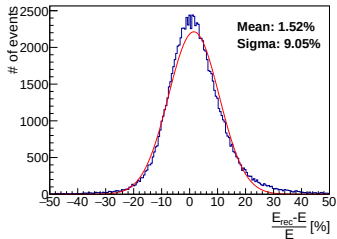
3 MeV



13 MeV



8 MeV



Relative reconstruction deviation of the kinetic energy of electron tracks with zero initial  $\theta, \varphi$ .



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- Several batches of tracks have been simulated for testing purposes.
  - $\theta \in [-17.1^\circ, 17.1^\circ]$ ,  $\varphi \in [-16.3^\circ, 16.3^\circ]$ ,  $E_k \in [3, 13]$  MeV
- The map of secondary electron positions and drift times has been generated.
- The map has been tested by the track reconstruction.
- Some of the systematic errors were corrected.
- The results suggest that:
  - Current energy resolution (fitted sigma) is 0.75 % for electrons and positrons under idealized conditions.
  - Multiple scattering is expected to reduce it significantly (up to  $\sim 11$  % for low energies), further investigation is needed.
  - OFTPC works well on a simulation level.



# Future Steps

- Complete the Geant4 simulation with multiple scattering for the full parameter grid
- Account for parasitic tracks caused by high energy secondary electrons
- Account for GEM in the simulation, charge distribution between pads
- Account for noise (denoising using a convolutional neural network — M. Gajdoš [17])
- Optimize Runge-Kutta integration fit with likelihood approach (instead of least squares) if needed
- Test different geometries
- Make a better simulation of magnetic and electric fields, compare with measurements



Thank you for your attention.



# References I

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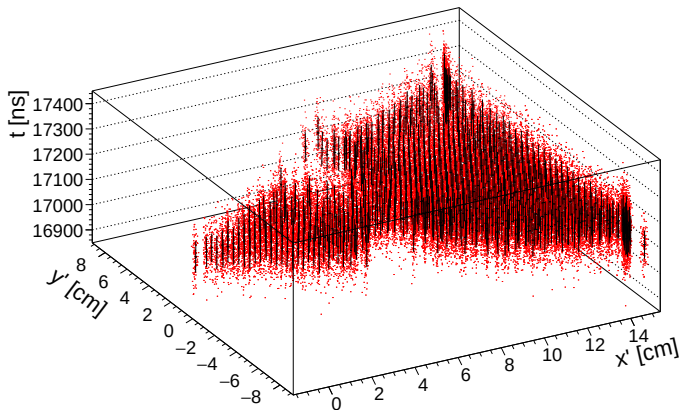




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# Ionization Electron Map Simulation



Mapping of the layer furthest from the OFTPC readout.

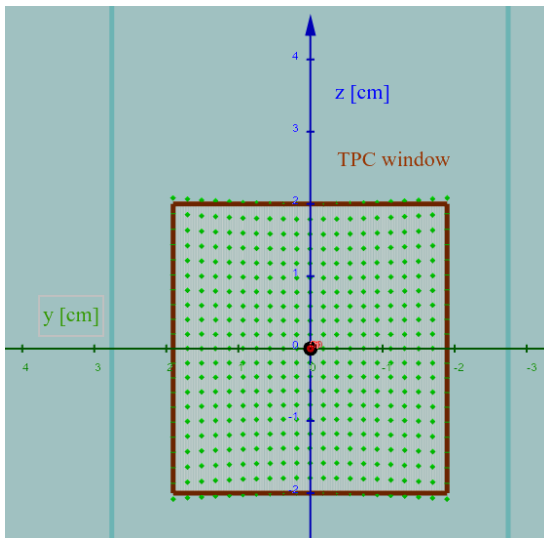


# Reconstruction Testing Summary

- 1 Ionization electron map simulation of secondary electrons in the entire volume of the OFTPC
- 2 Track(s) simulated using HEED and microscopic tracking
- 3 Counting the number of secondaries in each pad and time bin of the readout
- 4 Using the map to reconstruct the position of centers of pads for given centers of time bins
- 5 Fitting of the reconstructed points with circle and lines fit using least squares weighted by the number of secondaries in each point
- 6 Using the magnetic field in the middle of the track to get first energy estimate
- 7 Using the 4<sup>th</sup> order Runge-Kutta fit with least squares to refine the energy estimate
- 8 Correct systematic errors using a 3 parameter linear function



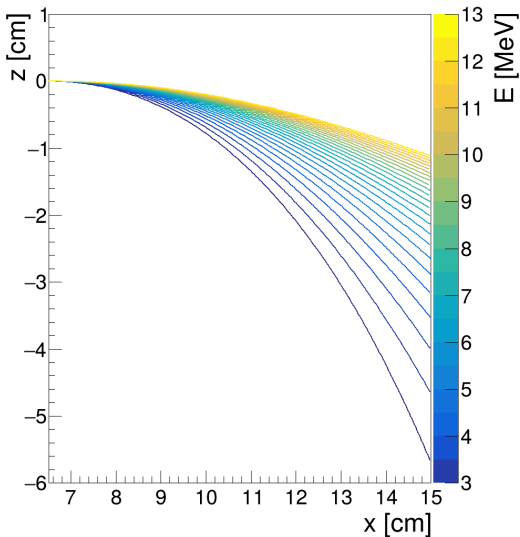
# Track Simulation – TPC Window



Initial points of tracks. Made with GeoGebra®.



# Runge-Kutta energy comparison



# Magnetic field in the XZ plane

